Table 1.2 Differences between prokaryotic and eukaryotic chromosomes

S.No	Prokaryotic chromosomes	Eukaryotic Chromosomes
1.	Many prokaryotes contain a single circular chromosome	Eukaryotes contain multiple linear chromosomes.
2.	Prokaryotic chromosomes are condensed in the nuceluoid via DNA supercoiling and the binding of various architectural proteins	Eukaryotic chromosomes are condensed via histones in a membrane bound nucleus
3.	Because prokaryotic DNA can interact with the cytoplasm transcription and translation occur simultaneously.	In eukaryotes, transcription occurs in the nucleus, and translation occurs in the cytoplasm.
4	Most prokaryotes contain only one copy of each gene (they are haploid) Nonessential prokaryotic genes are commonly encoded on extrachromosomal plasmids.	Most eukaryotes contain two copies of each gene (they are diploid)extrachromosomal plasmids are not commonly present in eukaryotes.
5	Prokaryotic genomes are compact and contain little repetitive DNA	Eukaryotes contain large amounts of noncoding and repetitive DNA.

The cell responds to this situation by controlling the movement of ions and water across the plasma membrane. Its ability to maintain a constant internal environment is called homeostasis. This is an important function of all cells, whether they are part of unicellular or multicellular organisms. One major role of homeostasis in animal cells is to cope with osmotic pressure by balancing the ionic composition so as to avoid the accumulation of water. To maintain homeostasis, ions and water may need to be moved into or out of a cell in a regulated manner.

The cell membrane regulates movement of water and ions. The difference in permeability to water and to ionic solutes has an important consequence in allowing osmotic pressure to develop across the membrane in response to differences in concentrations of dissolved substances on either side. Typically, the interior of the cell has higher potassium but lower sodium and calcium concentrations than that in the exterior. Sodium or potassium ions can diffuse less than the rate of water diffusion. So, there is a difference in ion concentrations on either side of the membrane. Water moves across to equalize the concentration of solutes on either side. If a cell had no mechanism to control solute levels, it would shrink or expand in response to osmotic pressure, whenever the concentration of solutes outside was greater than inside or *vice versa.*

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